

Unit 11, Lesson H1: Inscribing Polygons in Circles



Benchmark

MA.912.GR.5.4 Construct a regular polygon inscribed in a circle. Regular polygons are limited to triangles, quadrilaterals, and hexagons.

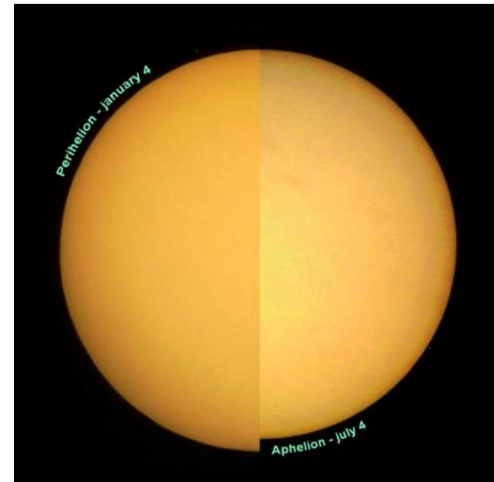


Learning Targets

- ☐ I can construct an equilateral triangle inscribed in a circle.
- ☐ I can construct a square inscribed in a circle.
- ☐ I can construct a regular hexagon inscribed in a circle.



11.H1.1 Warm-Up: Earth's Orbit



Since Earth's orbit is an ellipse, the Earth is slightly closer to the sun in January. The picture shows two photos taken by the SOHO satellite. The left side shows the sun on January 4, 2009 and the right side shows the sun on July 4, 2009.

A scientist measured the two images. She found that the diameter of the image for January was 72 millimeters and the image for July was 69 millimeters.

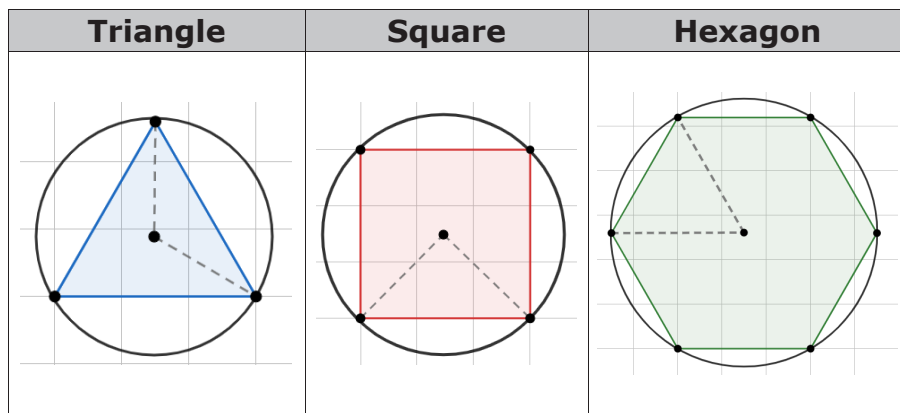
1. By what percentage did the diameter of the Sun change between January and July compared to its average diameter of January and July?
2. If the average distance to the Sun from Earth is 149,600,000 kilometers, how much closer is Earth to the Sun in July compared to January?



11.H1.2 Activity: Regular Polygons Inscribed in Circles

Work with your partner to answer each question. Use the larger copies of each polygon provided by your teacher.

1. Use a compass to draw a circumscribed circle for each polygon.
2. Mark the center of the circle that was drawn on the polygon. Draw a central angle that passes through two subsequent vertices, as shown.



3. Use a protractor to determine the measure of each central angle.

	Triangle	Square	Hexagon
Measure of Central Angle			

4. Make a conjecture about how to construct each regular polygon inscribed in a circle using a compass and a straight edge.

	Construction
Triangle	
Square	
Hexagon	

5. Ask another pair of partners for the conjectures they wrote. Listen to their conjectures and write a summary. *You are the only person who should write in the first three columns of the table.* Have the person initial next to your summary stating that your summary was accurate.

	Other Partner Pair’s Conjecture	Other Partner Pair’s Initials
Triangle		
Square		
Hexagon		

6. Review your conjectures with your partner and if necessary, make any changes.



11.H1.3 Activity: Card Sort

Sort the sets for each construction. Then, write the steps in the table.

	Construction Steps
Triangle Method 1	
Square Method 1	

**Hexagon
Method 1**

**Triangle
Method 2**

**Square
Method 2**

**Hexagon
Method 2**



11.H1.4 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about inscribing polygons in circles.

Unit 11, Lesson 3: Solving Problems Involving Triangles Inscribed in Circles



Benchmark

MA.912.GR.6.3 Solve mathematical problems involving triangles and quadrilaterals inscribed in a circle.



Learning Target

- ☐ I can solve mathematical problems involving triangles inscribed in a circle.